AMATH/PMATH 331: Applied Real Analysis Fall 2026

Assignment 3b; Solutions

---

[1] Let p be a monic polynomial of odd degree $2k+1$. This means that

$$p(x) = \sum_{j=0}^{2k+1} c_j x^j$$

with $c_j \in \mathbb{R}$ for $j \in \{0, 1, \ldots, 2k+1\}$ and $c_{2k+1} = 1$. Let $M = \max_{0 \le j \le 2k+1} |c_j| \ge 1$.

[a] Show that $p(2M) > 0$.

Hint: You will need to use the fact that $\sum_{j=0}^{2k} 2^j = \frac{1-2^{2k+1}}{1-2} = 2^{2k+1} - 1$.

[b] Show that $p(-2M) < 0$.

[c] Prove that there exists at least one point $x_o \in \mathbb{R}$ such that $p(x_o) = 0$.

Solution.

[a] By definition of M, we have $-M \le c_j \le M$ for all $0 \le j \le 2k+1$. Using this, and the formula for a geometric sum (namely, $\sum_{j=0}^{2k} 2^j = \frac{1-2^{2k+1}}{1-2} = 2^{2k+1} - 1$), we have

$$p(2M) = (2M)^{2k+1} + \sum_{j=0}^{2k} c_j (2M)^j \ge 2^{2k+1} M^{2k+1} + \sum_{j=0}^{2k} (-M) 2^j M^j = 2^{2k+1} M^{2k+1} - \sum_{j=0}^{2k} M^{j+1} 2^j$$

$$\ge 2^{2k+1} M^{2k+1} - \sum_{j=0}^{2k} M^{2k+1} 2^j = M^{2k+1}(2^{2k+1} - (2^{2k+1} - 1)) = M^{2k+1} > 0.$$

[b] Similarly we have

$$p(-2M) = (-2M)^{2k+1} + \sum_{j=0}^{2k} c_j (-2M)^j \le -2^{2k+1} M^{2k+1} + \sum_{j=0}^{2k} c_j 2^j M^j \le -2^{2k+1} M^{2k+1} + \sum_{j=0}^{2k} (M) 2^j M^j$$

$$\le -2^{2k+1} M^{2k+1} + \sum_{j=0}^{2k} 2^j M^{2k+1} = M^{2k+1}(-2^{2k+1} + (2^{2k+1} - 1)) = -M^{2k+1} < 0.$$

[c] Since p is a continuous real-valued function on $[-2M, 2M]$ and $p(-2M) < 0 < p(2M)$, by the Intermediate Value Theorem there exists $x_o \in \mathbb{R}$ such that $p(x_o) = 0$.

---

[2] Suppose that the surface of the Earth can be reasonably well-approximated to be a sphere $S = \{\mathbf{x} \in \mathbb{R}^3 : \|\mathbf{x}\| = R\}$ for some $R > 0$. It is intuitively clear, and not hard to prove, that S is *path-connected*. You do not have to show this.

Let $T \colon S \to \mathbb{R}$ be the temperature function on the surface of the Earth. That is, $T(\mathbf{x})$ is the temperature at the point $\mathbf{x} \in S$. Assume that T is a *continuous* function. Show that there must exist at least one point $\mathbf{p}$ on the surface of the Earth which is at exactly the same temperature at the *antipodal* point $-\mathbf{p}$. That is, prove that there exists $\mathbf{p} \in S$ such that $T(\mathbf{p}) = T(-\mathbf{p})$.

Hint: Consider the function $f \colon S \to \mathbb{R}$ given by $f(\mathbf{x}) = T(\mathbf{x}) - T(-\mathbf{x})$.

Solution. Since T is continuous, and the map $\mathbf{x} \mapsto -\mathbf{x}$ is continuous, we know that $f \colon S \to \mathbb{R}$ given by $f(\mathbf{x}) = T(\mathbf{x}) - T(-\mathbf{x})$ is continuous on S.

Let $\mathbf{q} \in S$ be arbitrary. If $f(\mathbf{q}) = 0$, then $T(\mathbf{q}) = T(-\mathbf{q})$, so we can take $\mathbf{p} = \mathbf{q}$ and we are done.

If $f(\mathbf{q}) \neq 0$, then it is either positive or negative. And

$$f(-\mathbf{q}) = T(-\mathbf{q}) - T(-(-\mathbf{q})) = -\big(T(\mathbf{q}) - T(-\mathbf{q})\big) = -f(\mathbf{q}).$$

Thus we have two points $\mathbf{q}, -\mathbf{q}$ in S with f being positive at one point and negative at the other point. We are told that S is path-connected. So by the Corollary to the Intermediate Value Theorem that we proved in class, there exists some $\mathbf{p} \in S$ with $f(\mathbf{p}) = 0$, so $T(\mathbf{p}) = T(-\mathbf{p})$.

---

[3] Let $f\colon (0,\infty) \to \mathbb{R}$ be given by $f(x) = \frac{1}{x}$. We proved in class that f is continuous on $(0,\infty)$.

[a] Show that f is *not* uniformly continuous on $A = (0,1)$.

[b] Show that f *is* uniformly continuous on $B = [1,\infty)$. [Note that B is *not compact*, so you cannot use the theorem from class. You need to do this directly.]

Solution.

[a] Take $\varepsilon = 1$. For any $\delta > 0$, choose $N \in \mathbb{N}$ such that $\frac{1}{N} < \delta$. Let $x = \frac{1}{N}$ and $y = \frac{1}{N+1}$, so $x, y \in A = (0,1)$. Then

$$|x - y| = \left|\frac{1}{N} - \frac{1}{N+1}\right| = \frac{1}{N(N+1)} < \delta,$$

and $|f(x) - f(y)| = |N - (N+1)| = 1 \geq \varepsilon$. Thus f is not uniformly continuous on $A = (0,1)$.

[b] Note that $x, y \in B = [1,\infty)$ means $x, y \geq 1$, so $\frac{1}{|xy|} \leq 1$. Given $\varepsilon > 0$, let $\delta = \varepsilon$. Then for every $x, y \in B$ with $|x - y| < \delta$, we have

$$|f(x) - f(y)| = \left|\frac{1}{x} - \frac{1}{y}\right| = \left|\frac{y - x}{xy}\right| = \frac{|x - y|}{|xy|} \leq |x - y| < \delta = \varepsilon.$$

Thus f is uniformly continuous on $B = [1,\infty)$.

---

[4] Fix a subset $E \subseteq \mathbb{R}^n$. Let $f, g\colon E \to \mathbb{R}^m$.

[a] Suppose that f and g are both uniformly continuous on E. Let $\lambda, \mu \in \mathbb{R}$. Show that $\lambda f + \mu g$ is also uniformly continuous on E.

[This shows that the space of functions from E to $\mathbb{R}^m$ which are uniformly continuous on E is a linear subspace of the vector space of all continuous functions from E to $\mathbb{R}^m$.]

[b] Let $m = 1$, so $fg\colon E \to \mathbb{R}$ is defined. Suppose that f and g are both uniformly continuous on E, and that they are both *bounded*. Show fg is uniformly continuous on E.

[This shows that the space of *bounded* uniformly continuous functions from E to $\mathbb{R}$ is closed under multiplication. In the language of abstract algebra, it is a *subring* of the ring of continuous functions from E to $\mathbb{R}$.]

[c] Show that the assumption that f, g are bounded is necessary for part [b]. That is, find examples of $f, g\colon E \subseteq \mathbb{R}^n \to \mathbb{R}$ which are uniformly continuous on E, and such that at least one of f, g is unbounded, and such that fg is not uniformly continuous on E. [Take $n = 1$ for simplicity.]

Solution.

[a] Let $\varepsilon > 0$. There exists $\delta_1 > 0$ such that

$$\mathbf{x}, \mathbf{y} \in E \text{ and } \|\mathbf{x} - \mathbf{y}\| < \delta_1 \text{ implies } \|f(\mathbf{x}) - f(\mathbf{y})\| < \tfrac{1}{2(1+|\lambda|)}\varepsilon.$$

There exists $\delta_2 > 0$ and such that

$$\mathbf{x}, \mathbf{y} \in E \text{ and } \|\mathbf{x} - \mathbf{y}\| < \delta_2 \text{ implies } \|g(\mathbf{x}) - g(\mathbf{y})\| < \tfrac{1}{2(1+|\mu|)}\varepsilon.$$

Let $\delta = \min\{\delta_1, \delta_2\} > 0$. Then for $\mathbf{x}, \mathbf{y} \in E$ with $\|\mathbf{x} - \mathbf{y}\| < \delta$, we have

$$\begin{aligned}
\|(\lambda f + \mu g)(\mathbf{x}) - (\lambda f + \mu g)(\mathbf{y})\| &= \|\lambda\big(f(\mathbf{x}) - f(\mathbf{y})\big) + \mu\big(g(\mathbf{x}) - g(\mathbf{y})\big)\| \\
&\leq |\lambda|\,\|f(\mathbf{x}) - f(\mathbf{y})\| + |\mu|\,\|g(\mathbf{x}) - g(\mathbf{y})\| \\
&< \tfrac{|\lambda|}{2(1+|\lambda|)}\varepsilon + \tfrac{|\mu|}{2(1+|\mu|)}\varepsilon < \varepsilon.
\end{aligned}$$

Thus $\lambda f + \mu g$ is uniformly continuous on E.

[b] Since f, g are both bounded, there exists $C > 0$ such that $|f(\mathbf{x})| \leq C$ and $|g(\mathbf{x})| \leq C$ for all $\mathbf{x} \in E$. Let $\varepsilon > 0$. By taking the minimum of a $\delta_1 > 0$ and a $\delta_2 > 0$ as in part [a], we can find $\delta > 0$ such that

$$\mathbf{x}, \mathbf{y} \in E \text{ and } \|\mathbf{x} - \mathbf{y}\| < \delta \text{ implies that } |f(\mathbf{x}) - f(\mathbf{y})| < \tfrac{\varepsilon}{2C} \text{ and } |g(\mathbf{x}) - g(\mathbf{y})| < \tfrac{\varepsilon}{2C}.$$

Then for $\mathbf{x}, \mathbf{y} \in E$ with $\|\mathbf{x} - \mathbf{y}\| < \delta$, we have

$$\begin{aligned} |(fg)(\mathbf{x}) - (fg)(\mathbf{y})| &= |f(\mathbf{x})g(\mathbf{x}) - f(\mathbf{y})g(\mathbf{y})| \\ &= |f(\mathbf{x})g(\mathbf{x}) - f(\mathbf{x})g(\mathbf{y}) + f(\mathbf{x})g(\mathbf{y}) - f(\mathbf{y})g(\mathbf{y})\| \\ &= |f(\mathbf{x}|\,|g(\mathbf{x}) - g(\mathbf{y})| + |g(\mathbf{y})|\,|f(\mathbf{x}) - f(\mathbf{y})| \\ &\leq C\tfrac{\varepsilon}{2C} + C\tfrac{\varepsilon}{2C} = \varepsilon. \end{aligned}$$

Thus fg is uniformly continuous on E.

[c] Take $E = \mathbb{R}$, and $f, g \colon \mathbb{R} \to \mathbb{R}$ both given by $f(x) = g(x) = x$. Then f, g are linear and thus uniformly continuous, and also unbounded. But then $(fg)(x) = x^2$ which is not uniformly continuous on $\mathbb{R}$.

---

[5] Suppose that $f \colon \mathbb{R}^n \to \mathbb{R}$ is continuous on $\mathbb{R}^n$ and that f *goes to zero at infinity* in the sense of problem [5] from Assignment #3a. Prove that f is uniformly continuous on $\mathbb{R}^n$.

Solution. Let $\varepsilon > 0$. Since f goes to zero at infinity, there exists $M > 0$ such that $\|\mathbf{x}\| > M$ implies that $|f(\mathbf{x})| < \frac{\varepsilon}{2}$. On the compact set $\overline{B_{2M}(\mathbf{0})}$, we know that the continuous function f is uniformly continuous. Thus there exists $\tilde{\delta} > 0$ such that $\mathbf{x}, \mathbf{y} \in \overline{B_{2M}(\mathbf{0})}$ and $\|\mathbf{x} - \mathbf{y}\| < \tilde{\delta}$ implies that $|f(\mathbf{x}) - f(\mathbf{y})| < \varepsilon$. Let $\delta = \min\{\tilde{\delta}, M\}$. Suppose $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ with $\|\mathbf{x} - \mathbf{y}\| < \delta$. There are three cases.

Case one: Both $\|\mathbf{x}\|, \|\mathbf{y}\| \leq 2M$, so $\mathbf{x}, \mathbf{y} \in \overline{B_{2M}(\mathbf{0})}$. Then, since $\delta \leq \tilde{\delta}$, we have $|f(\mathbf{x}) - f(\mathbf{y})| < \varepsilon$.

Case two: Both $\|\mathbf{x}\|, \|\mathbf{y}\| > M$. Then $|f(\mathbf{x})|, |f(\mathbf{y})| < \frac{\varepsilon}{2}$, so $|f(\mathbf{x}) - f(\mathbf{y})| \leq |f(\mathbf{x})| + |f(\mathbf{y})| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$.

Case three: The only remaining case, up to swapping the roles of $\mathbf{x}, \mathbf{y}$, is that $\|\mathbf{x}\| < M$ and $\|\mathbf{y}\| > 2M$. But this cannot happen, because

$$\|\mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y} - \mathbf{x}\| < M + \delta \leq 2M$$

since $\delta \leq M$.
